



User Manual

Mobile Application

Designed for Social Workers & Adopted Children

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1. Introduction

Haden is an Artificial intelligence (AI)-powered emotional support and monitoring system designed to assist social workers in monitoring children's emotional and psychological wellbeing.

Traditional monitoring methods mainly rely on scheduled visits every few months, which may delay the detection of emotional distress or behavioral issues. Haden aims to improve this process through intelligent and continuous monitoring technologies.

The system combines the following artificial intelligence technologies:

- Facial emotion detection
- Intelligent chatbot interaction
- Automated report generation

The application provides organized reports and alerts to help social workers make informed decisions and support children more effectively.

2. About Haden

Haden was developed to support adopted children and improve post-adoption follow-up processes using artificial intelligence technologies.

The system focuses on helping social workers monitor emotional wellbeing through interactive sessions between the child and an AI companion.

During each session, Haden analyzes facial expressions to provide emotional insights and generate detailed reports.

Important: The system is designed as a supportive monitoring tool and does not replace professional psychological evaluation or human supervision.

3. Intended Users

Haden is intended for the following users:

- **Adopted children:** between the ages of **6 and 12**
- **Social workers:** authorized users with full dashboard access

4. System Requirements

Before using Haden, ensure your device meets the following requirements:

Requirement	Details
Internet Connection	Required for all features
Device	Smartphone (Android)
Camera	Required for facial emotion detection
Microphone	Required for AI companion interaction

Note: If permissions are denied, all features may not work correctly. Grant all required permissions upon first launch.

5. Main Features

Feature	Description
AI Companion (Noor)	An intelligent chatbot that interacts with the child during sessions called Noor.
Facial Emotion Detection	Analyzes facial expressions in real-time to identify emotions
Session Reports	Automatically generated reports summarizing each session
Alerts & Notifications	Send alerts to social workers when concerns are detected
Monitoring Dashboard	A centralized view for social workers to review all session data

6. How Haden Works in General

A typical session follows these steps:

1. **Start Session:** The child initiates a new session.
2. **Activate Camera and Microphone:** Permissions are confirmed and hardware is enabled.
3. **Begin Interaction:** The child interacts with the AI companion through conversation.
4. **Analyze Facial Expressions:** The system continuously analyzes the child's emotions.
5. **Monitor Results:** The social worker can observe session data in real time.
6. **Generate Session Report:** A detailed report is automatically created at session end.
7. **Send Alerts if Needed:** If emotional concerns are detected, alerts are dispatched.
8. **End Session:** The session is closed and data is saved.

7. Recommended Conditions

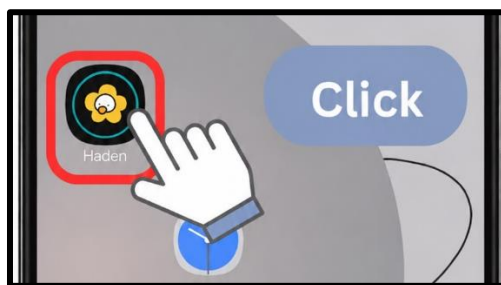
For the best experience and most accurate results, ensure the following conditions:

- Good lighting: Ensure the child's face is clearly visible to the camera.
- Stable internet connection: A reliable connection is required for all AI features.
- Quiet environment: Minimize background noise to improve AI companion interaction.

8. Getting Started

The user begins by opening the Haden application through the app icon on the device. After launching the application, the splash screen appears while the system loads. The user is then directed to the login page, where they enter their ID and password. Once the login information is verified, the system automatically identifies the user type and redirects them to the appropriate interface. Adopted children are automatically directed to the child interface, while social workers are automatically directed to the social

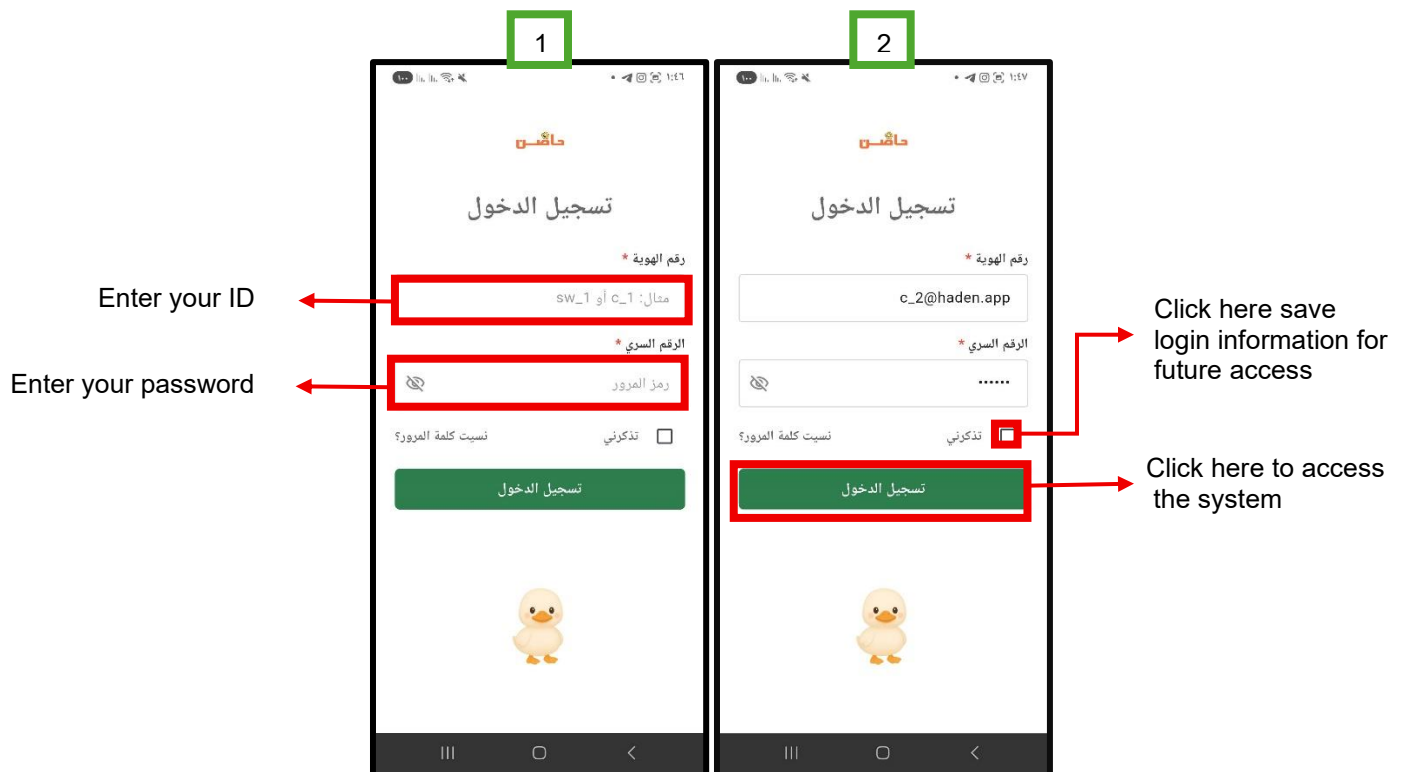
1. To get started with Haden as a child; a member of the family or the same child will click on the app icon:



2. After clicking, an -application intro- will appear on the screen while the application is loading:



3. After the splash screen, a log-in page will appear on the screen:



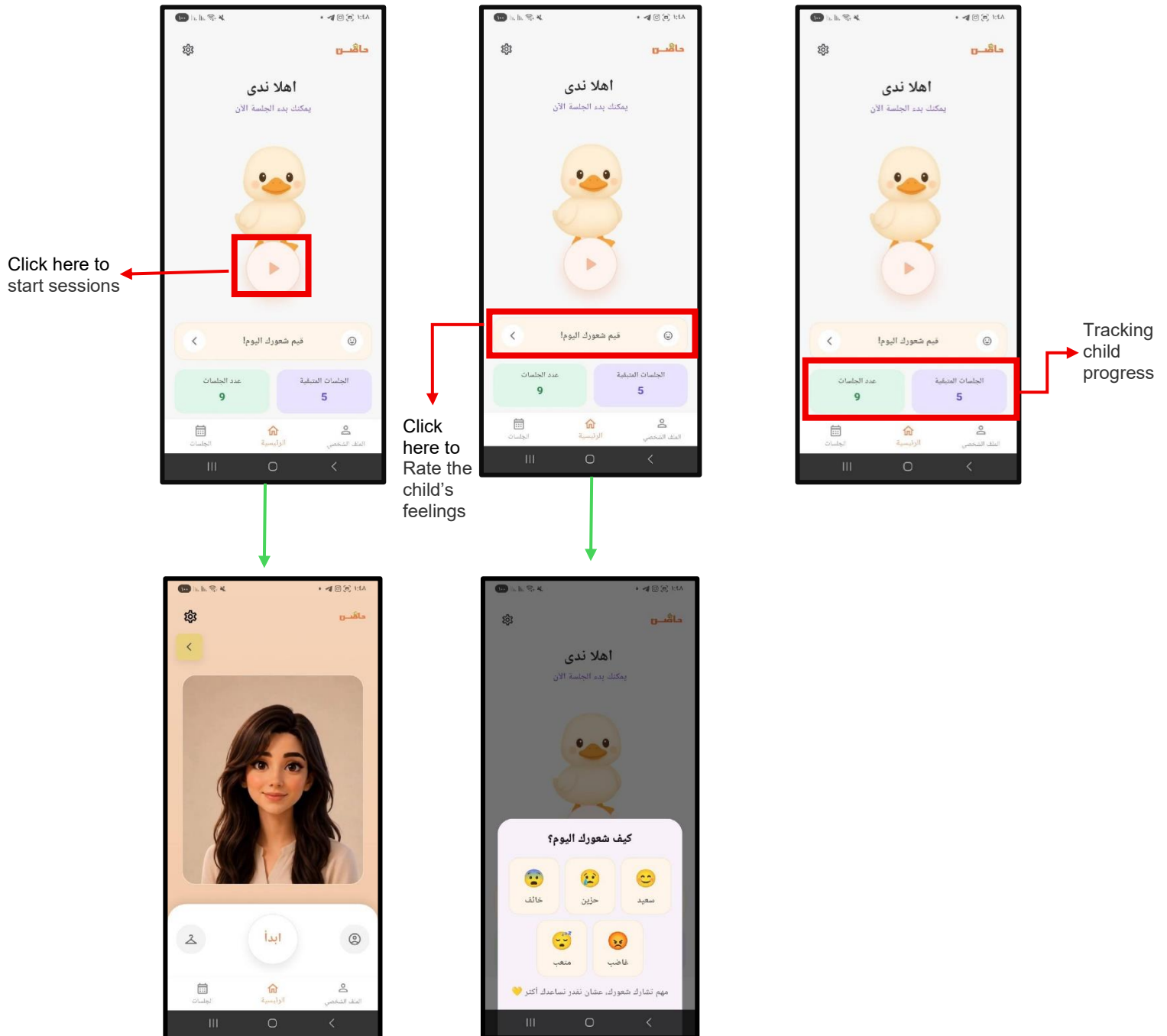
Based on the user type automatically redirects them to the appropriate interface.

8.1 Child Interface

The child interfaces are designed to provide a simple and friendly experience. Children can: view upcoming sessions, join a session, Interact with the AI companion

- The child's main screen allows the user to:

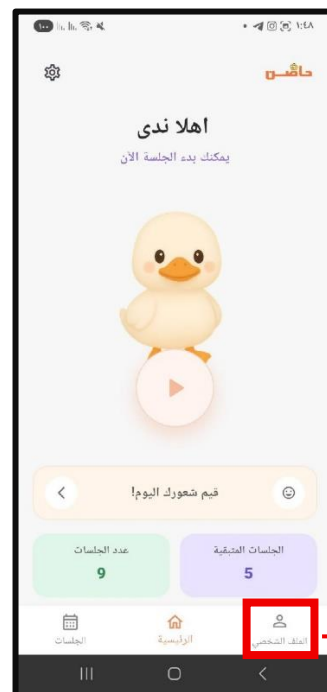
start a session	view remaining sessions and track completed sessions	child can select their current emotion
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access the child profile

view remaining sessions and
track completed sessions

access the child profile

Click here
to log-outClick here for
the session
scheduleClick here
for the child
profile

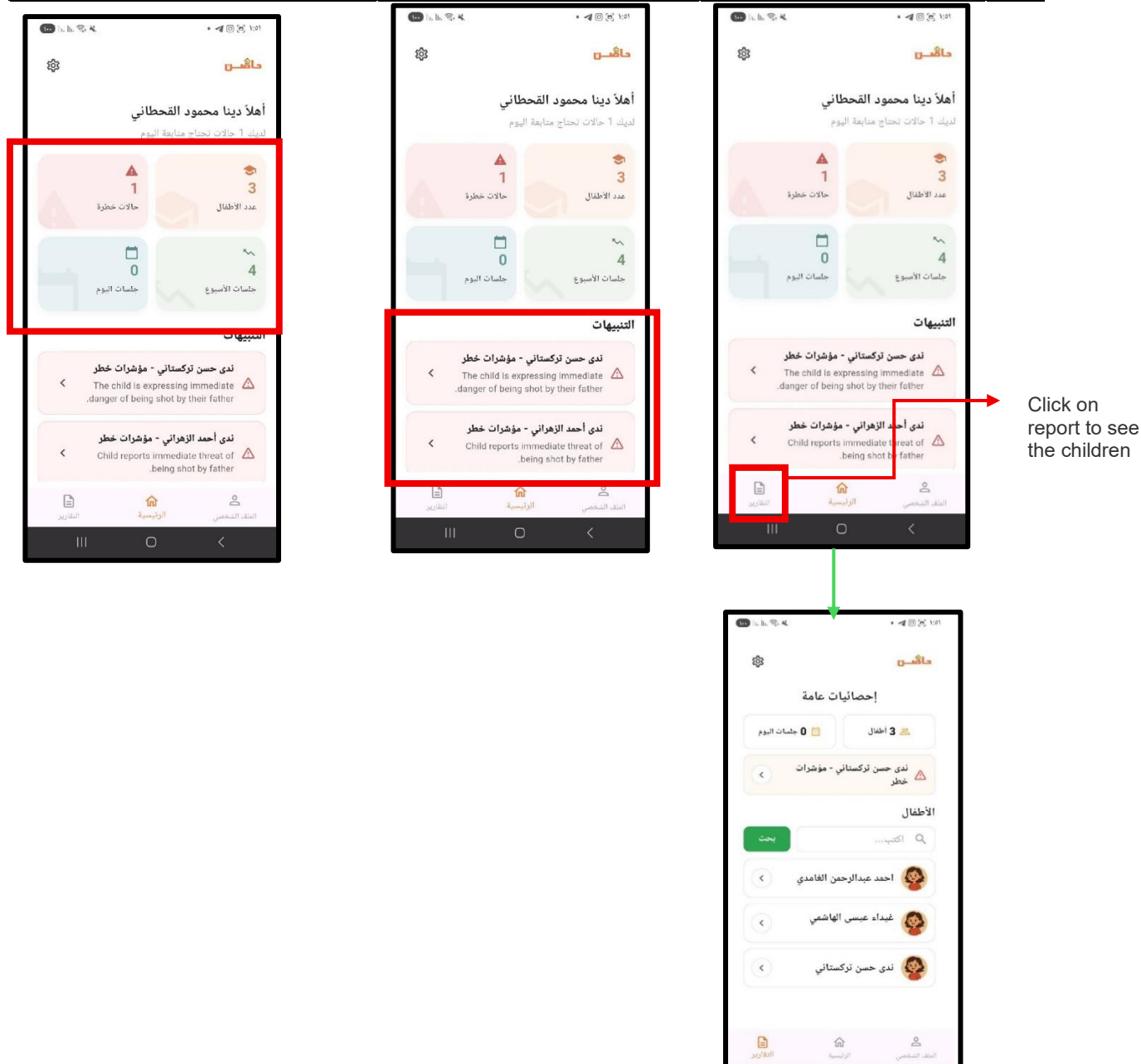
Child and parents cannot access reports, alerts, or the monitoring dashboard.

8.2 Social Worker Interfaces

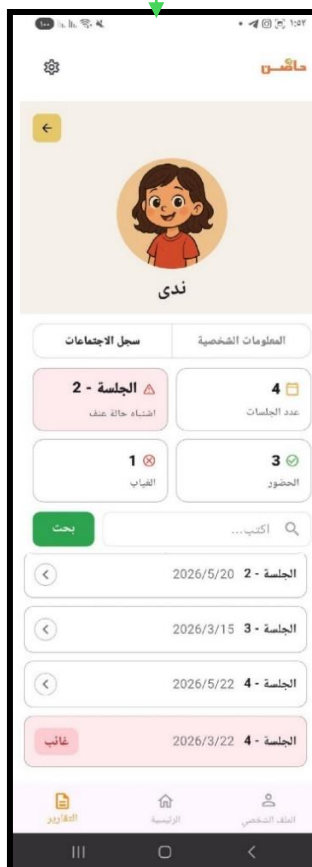
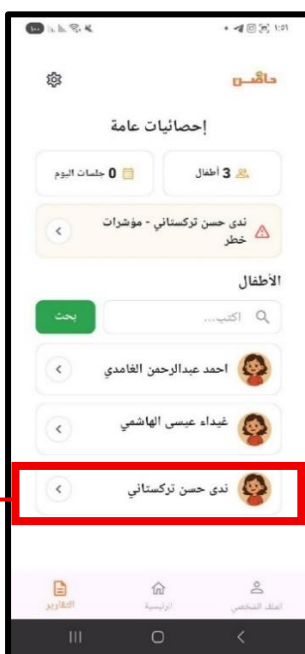
The social worker is an authorized user with access to information about each child under their supervision and all data in theme dashboard. The dashboard includes session reports, alerts and notifications, session history and emotion data detected through the child's facial expressions

- The social worker main screen allows them to:

Displays overall statistics, number of children under supervision, and number of cases that require immediate attention.	Displays overall active alerts	Click on report to see the children Under supervision
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Select a child to view details	Displays detailed session history	social worker profile
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- After clicking on the meeting log section and scrolling down, the user can:

view the session history. Selecting a session displays its detailed information.

Displays emotional analysis, AI alerts, recommendations, risk indicators, and session details.

Click on the session you want to review



scroll down



9. System Re-Run Instructions

This section is responsible for re-running the project, you can find the structure and the steps required to set up and launch the full system.

The full system contains:

- **AI_Modules**: Contains all artificial intelligence components of the system.
- **flutterProject**: Contains the Flutter mobile application and its backend servers.

9.1 AI_Modules Folder

A. The first step is downloading the **AI_Modules Folder** to your laptop or PC:

AI_Modules	File folder	5/28/2026 6:29 AM
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B. Inside AI_Modules, there are two components and you can try each part of the system separately:

chatbot_Noor	File folder	5/28/2026 5:30 AM
Face Emotion Detection	File folder	5/28/2026 6:21 AM

1. chatbot_Noor

This folder contains the AI companion Noor. Inside, you will find the **Noor_Dialog.py** file which is the main script for the chatbot. This file is for trying the of the system by chatting the AI companion by voice.

[Note: This file must be opened in Visual Studio Code or something similar (not google colab!)]

Noor_Dialog.py	Python Source File	15 KB	No	43 KB	66%	5/28/2026 5:25 AM
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To configure Noor for your environment, you may update the following settings inside this file to make Noor interact with you by the mic (Noor is the name of the AI companion):

- **API Channel**: Update the channel configuration to match your microphone (mic) settings.
- **Other configurable parameters**: Any other connection or environment specific settings referenced in the script must also be updated accordingly based on your device.

[Note: Noor interacts with the child using voice only. No video is generated during conversations.]

2. Face Emotion Detection

You can use face emotion detection alone. This folder contains five components for the facial emotion detection model:

Name	Type	Compressed size	Password p...	Size	Ratio	Date modified
Ready_weights	File folder					5/28/2026 5:41 AM
FED_preprocessing.ipynb	Jupyter Source File	2,178 KB	No	2,918 KB	26%	5/28/2026 6:16 AM
FED_training.ipynb	Jupyter Source File	1,208 KB	No	2,023 KB	41%	5/28/2026 5:58 AM
realTime.py	Python Source File	4 KB	No	9 KB	64%	5/28/2026 6:21 AM

- **Sampels_Dataset:** Private Training Dataset The face model was trained using the LIRIS-CSE dataset as the primary dataset. In addition, a set of videos recorded by our team was used to support the training process, particularly for improving class balance and enhancing the model's performance in recognizing children's facial expressions. Since these videos contain private data, they cannot be shared publicly. Therefore, only a small sample has been provided.
- **FED_preprocessing.ipynb:** A Jupyter Notebook documenting all preprocessing steps applied to the dataset before training.
- **FED_training.ipynb:** A Jupyter Notebook documenting the full model training process, including all steps and configurations.

*Note: These two files **can** be opened in Visual Studio Code, google colab or something similar*

- **realTime.py:** The real-time emotion detection script. This file must be opened in **Visual Studio Code** and must be placed in the same folder as the **Ready_weights** folder containing the pretrained model weights. No retraining or re-preprocessing is needed the weights are ready to use.

Note: This file must be opened in Visual Studio Code or something similar (not google colab!)

If you want to use the full system as an app and connected units, you will need this part:

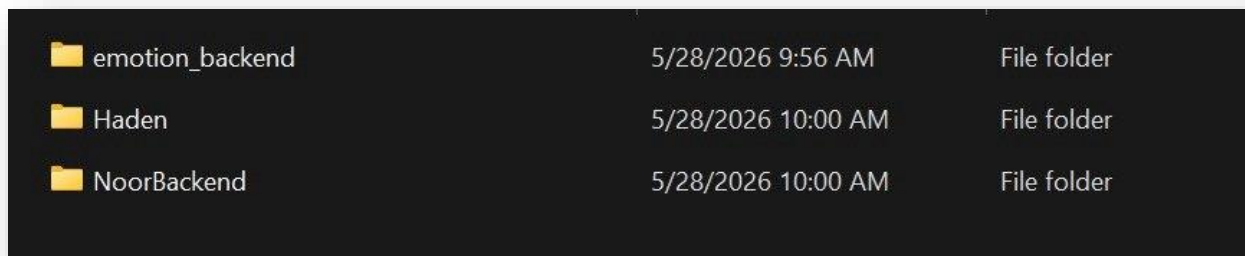
Note: We use Flutter for building the app

The Flutter project files are distributed via GitHub. Download the source code from the project repository using the GitHub link. [Click here](#)

Once downloaded, ensure that the two backend folders (**NoorBackend** and **emotion_backend**) are present in the same directory as the **Haden** Flutter project folder.

9.2 flutterProject Folder

Inside flutterProject, there are three sub-folders:



emotion_backend	5/28/2026 9:56 AM	File folder
Haden	5/28/2026 10:00 AM	File folder
NoorBackend	5/28/2026 10:00 AM	File folder

- **emotion_backend**: The backend server that powers the facial emotion detection feature within the application.
- **NoorBackend**: The backend server for the AI companion Noor, which handles chatbot interactions and video session management.
- **Haden**: The main Flutter application project.

9.2.1 Running the System

Three terminal windows are required to run the full system. Follow the steps below in order.

Terminal 1 — Start NoorBackend

Navigate to the NoorBackend folder, then run the following commands:

```
cd NoorBackend
(Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (&
c:\Users\amjaa\ai_env\Scripts\Activate.ps1)
python -m uvicorn noor_api:app_api --host 0.0.0.0 --port 8001
```

Terminal 2 — Start emotion_backend

Open a new terminal, navigate to the emotion_backend folder, then run:

```
cd emotion_backend
(Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (&
c:\Users\amjaa\ai_env\Scripts\Activate.ps1)
python -m uvicorn emotion_api:app --host 0.0.0.0 --port 8002
```

Terminal 3 — Run the Flutter Application

Open a third terminal, navigate to the Haden folder, then run:

```
cd Haden
flutter run
```

Important: All three terminals must remain open and run simultaneously for the application to function correctly.

10. Troubleshooting

If you experience any issues, refer to the table below:

Problem	Solution
App does not open	Restart the application and try again.
Camera not working	Go to device Settings and allow camera permission for Haden.
Microphone not working	Go to device Settings and allow microphone permission for Haden.
Slow performance	Check your internet connection and ensure sufficient signal strength.
Emotion analysis inaccurate	Improve lighting conditions and ensure the child's face is fully visible.

11. Frequently Asked Questions (FAQ)

Question	Answer
Who can use the Haden application?	Adopted children and social workers.
How do I start a session?	Press the start session button on the home screen for the child.
What does the emotion analysis feature do?	It analyzes the child's facial emotions during sessions.
Can the social worker view session reports?	Yes, detailed reports and session history are available.
What are AI alerts?	AI alerts notify social workers about possible risk indicators.
What happens after login?	The system automatically opens the correct interface based on the user type.
Does the application save login information for future visits?	Yes, by enabling the "Remember Me" option.

Will the AI companion tell the child that they are adopted?	No, the system avoids answering sensitive adoption related questions.
Are the AI analysis results reviewed by a social worker?	Yes, all AI-generated analyses and recommendations are reviewed by a social worker.
Can the AI system make decisions without human supervision?	No, the system supports social workers and does not replace professional judgment.
What happens if camera or microphone permissions are denied?	Camera and microphone permissions must be accepted. If the permissions are denied, a social worker will contact the parents for follow up.
What happens if parents stay in the room during the session?	These features are not applicable now.
What if my child feels shy or does not want to talk to the AI companion?	The AI companion will begin with a warm-up session using simple games and friendly interaction to help the child feel comfortable before gradually starting deeper conversations.
Can the application monitor children who are 4 or 5 years old?	No, Haden is designed for children between the ages of 6 and 12 years old.
Can parents view the child's session results and conversations with the AI companion?	No, session results and conversations are only accessible to authorized social workers.
Can the child use the application without internet connection?	No, internet connection is required for the AI features and session monitoring.
Can sessions be rescheduled?	Yes, social workers can manage and reschedule sessions when needed.
Is the child's information kept private?	Yes, all information and session data are securely protected.
Can social workers track previous sessions?	Yes, session history and reports are available for review.
Can multiple children use the same account?	No, each child has a separate profile and monitoring record.

— End of User Manual —